RESPONSIBLE MACHINE LEARNING AUDIT

Home Mortgage Lending Model Audit

COURSE	DNSC 6330 Capstone Presentation
INSTITUTION	George Washington University School of Business

HMDA lending model evaluated for predictive performance, disparate impact, residual risk, and deployment defensibility.

OPTIMIZATION OBJECTIVE

Metric, threshold, and trade-off

METRIC

F1 = 0.8861

Validation-optimized balance of precision + recall

THRESHOLD

0.2575

Operational approval cutoff

RANKING

AUC 0.8127

Test set ranking quality

DECISION RULE

Approve if **P(approval) ≥ 0.2575**. Rejected otherwise — this is where false denials become the key ethical harm.

METRIC & THRESHOLD

Specification

CRITERION	SPECIFICATION
Metric	F1 Score ≥ 0.886
Threshold	0.2575 (cutoff probability for approval prediction)
Optimization Approach	F1-maximization on validation set (balanced precision-recall)
Decision Rule	IF P(approval) ≥ 0.2575 THEN approve; ELSE deny

WHY THIS THRESHOLD?

The validation set was sorted by predicted probability, and the threshold that maximizes F1 was selected. This differs from the common 0.5 default because F1 prioritizes finding the sweet spot between false positives (lender losses) and false negatives (applicant exclusion).

BUSINESS RATIONALE

Two sides of the F1 balance

FOR LENDERS

Mortgage lending requires **minimizing credit risk** (false positives) — lending to unqualified borrowers who default is costly

F1 helps balance this by maintaining **high precision** (91% of approved applicants actually repay)

FOR APPLICANTS & FAIRNESS

The U.S. mortgage market is critical infrastructure for wealth-building; **wrongful denials = long-term economic exclusion**

HMDA regulations (12 CFR 1003) require lenders to **avoid discrimination** based on protected attributes

THIS MODEL CONNECTS LENDING DECISIONS TO APPLICANT CONSEQUENCES

Approved (~81% of applicants)

Access to credit, homeownership pathway, generational wealth

Denied (~19% of applicants)

Exclusion from mortgage market, forced into subprime/rental markets, household financial instability

BUSINESS CONTEXT

Mortgage decisions are high-stakes ML decisions

The harm is not only statistical: lending errors affect access to housing, credit, and wealth creation.

HMDA

Public mortgage-lending context

RACE + SEX

Protected groups audited

4/5 RULE

AIR < 0.80 red flag

A WRONG DENIAL CAN MEAN

- Delayed homeownership
- Higher borrowing costs
- Reduced credit access
- Lost wealth-building opportunity
- Regulatory and reputational risk

AUDIT QUESTION

Can this model support mortgage decisions without creating unjustified disparate impact — and can it be defended to community, legal, and fintech stakeholders?

THE CORE TENSION

Accuracy is not fairness

F1 optimization balances *lender safety* and *overall accuracy*, but **does not ensure fairness** . A model can be accurate on average while systematically harming specific groups.

TRADE-OFF ANALYSIS

Who pays the cost, and what the metrics show

GROUP	COST OF MODEL	METRIC EVIDENCE
Low-risk applicants (entire population)	**Wrongful denials** → excluded from credit market	129,152 wrongfully denied in low-risk tier; 42% false negative rate
Racial minorities (Black/African American, Native Hawaiian, "Free Form Text Only")	**Disparate impact** → systemic exclusion by race	38.2% approval gap: Joint (80.3%) vs. Free Form Text Only (42.0%); Black (63.3%) vs. White (78.3%)
Female applicants	**Gender-based exclusion** → compounding disadvantage	11.2% lower approval rate (Women 70.6% vs. Men 74.3%); 33% false negative rate
Lenders	**Credit risk** from false positives	~30K wrongfully approved in high-risk tier; potential loan losses

AUDIENCE & FRAMING

Who is affected, and what counts as failure

STAKEHOLDERS & HARM ASYMMETRY

Applicants

Bear the dominant harm. Protected-class applicants disproportionately so.
A wrongful denial blocks credit for a qualified borrower.

Lender

Carries institutional risk: bad loans approved, regulatory exposure, reputational damage from disparate-impact findings.

Regulators

CFPB, HUD, OCC examine the lender. Fair-lending exam cycles focus on FNR by protected class.

Brokers / vendors

Influence input integrity. Income/DTI gaming and census-data corruption are documented threat vectors.

TAKEAWAY

A **false negative** — denying a qualified applicant — is the worst outcome and the metric fair-lending exam cycles focus on. We lead the audit with that harm metric.

MODEL OVERVIEW

How the prediction system is audited

INPUT	FEATURES	MODEL	DECISION RULE	AUDIT OUTPUTS
HMDA applications	**30 model inputs**	**GBM model**	**Approve if**	**AIR, ME, SMD**
Processed test set 1,190,080 rows	Non-protected + operational	Approval probability	Score meets threshold	FPR/FNR + Drift

PERFORMANCE EVIDENCE

GBM selected because it performs best overall

The performance claim is specific, measured, and tied to the model choice.

ROC-AUC

0.8127

Stronger ranking than Logistic Regression

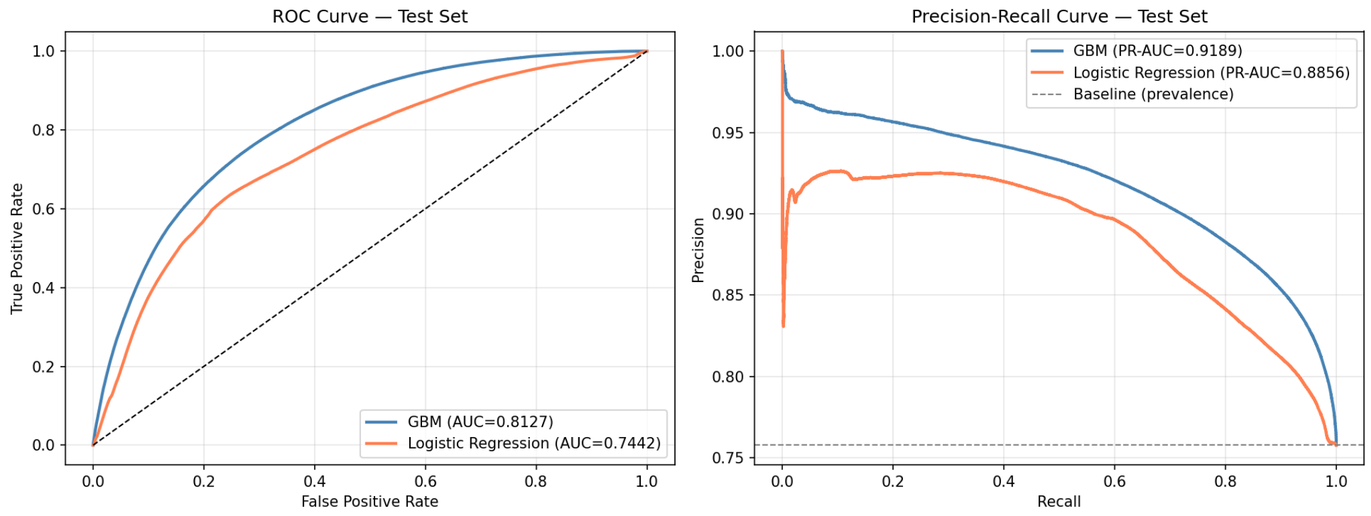
PR-AUC

0.9189

Strong precision-recall performance

NOT ENOUGH

Aggregate performance still requires subgroup audit



FOUR FINDINGS

What the audit found

A strong model can still create legally and ethically meaningful subgroup harm.

1

The GBM performs well overall

ROC-AUC = 0.8127 and PR-AUC = 0.9189; chosen over Logistic Regression.

2

Average performance hides harm

At threshold 0.2575, 39,224 approved applicants are predicted as denied.

3

Fair-lending red flags exist

AIR < 0.80 for three race groups under the baseline fairness audit.

4

Deployment is conditional

Threshold mitigation improves AIR, but residual risk requires governance and monitoring.

KNOWN FAILURE MODES

Where the model breaks down, who is harmed, and why

1 Wrongful denials

EVIDENCE

39,224 false denials at threshold 0.2575; low-risk tier contains 129,152 approved borrowers predicted denied.

HARM

Creditworthy applicants face delayed homeownership and reduced financial mobility.

2 Disparate impact

EVIDENCE

AIR < 0.80 for Free Form Text Only, Native Hawaiian or Other Pacific Islander, and American Indian or Alaska Native applicants.

HARM

Protected groups receive lower approval access relative to White applicants.

3 Threshold sensitivity

EVIDENCE

Subgroup AIR changes materially when the approval cutoff moves from 0.50 to 0.20.

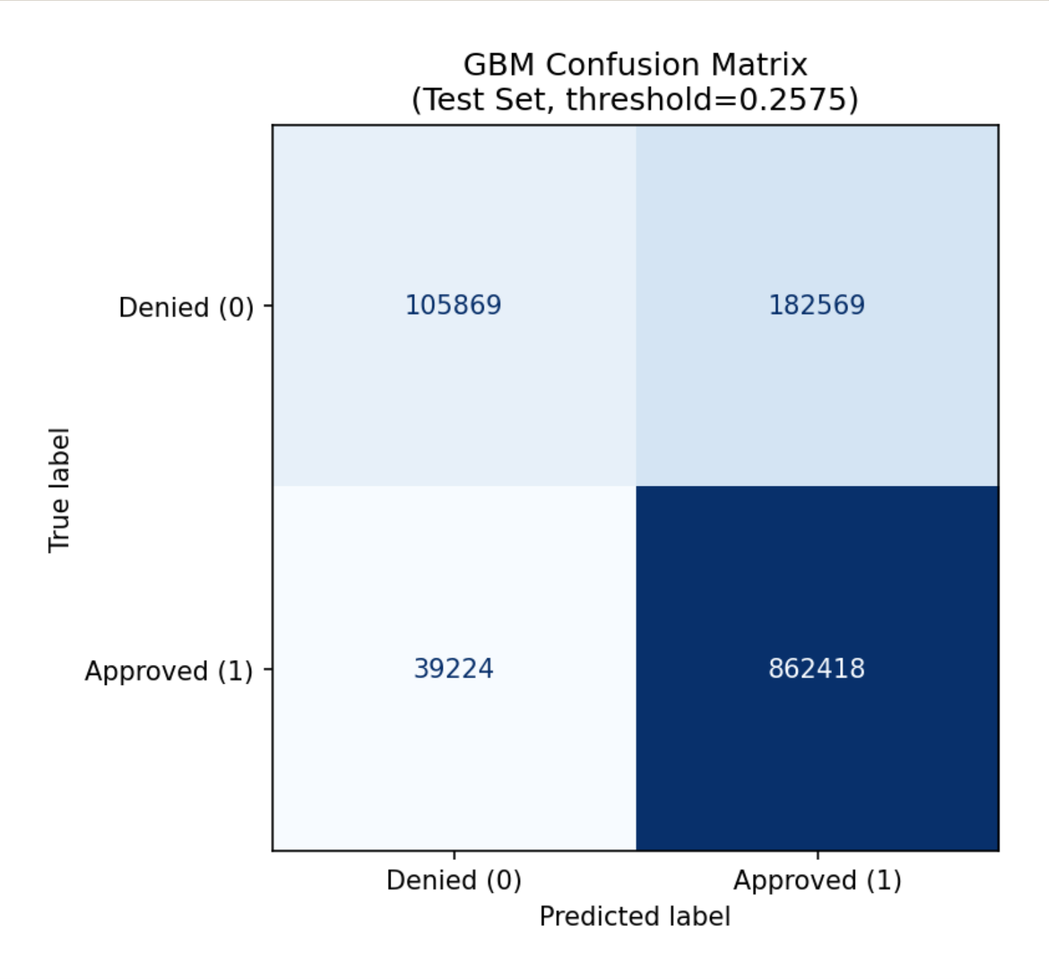
HARM

Fairness depends on a governance decision, not only model performance.

EVIDENCE

CLASSIFICATION OUTCOMES · AT THRESHOLD = 0.2575

False denials are the central ethical harm



39,224

false denials: actually approved, predicted denied

182,569

false approvals: actually denied, predicted approved

862,418

true approvals

105,869

true denials

SUBGROUP FAIRNESS EVALUATION

AIR flags the most important fair-lending risk

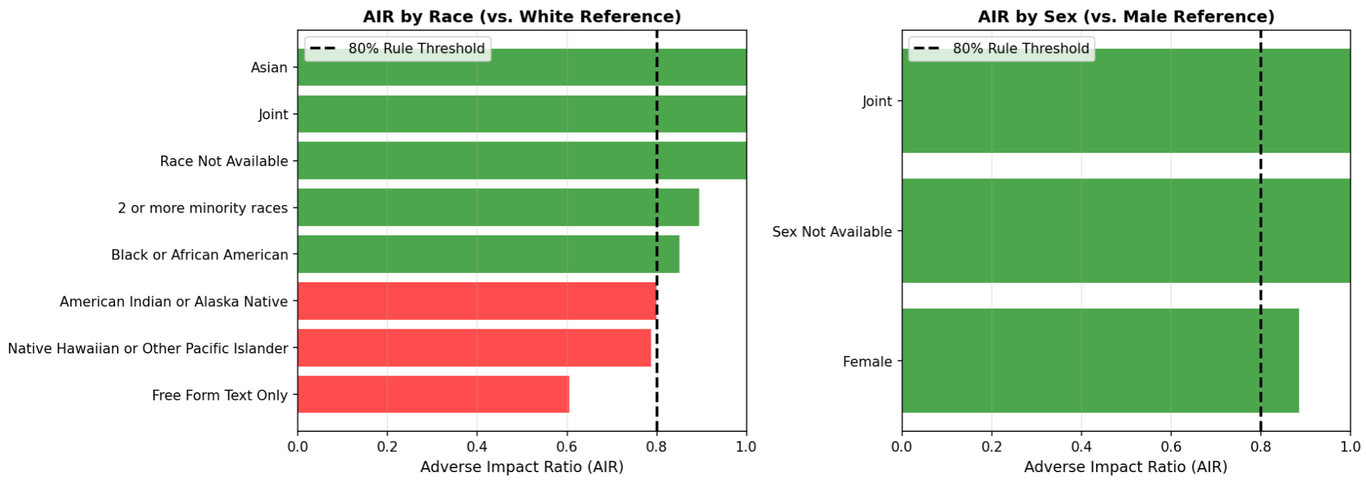
AIR = group approval rate ÷ White approval rate

AIR < 0.80 is a four-fifths-rule red flag.

Lowest AIR = 0.606 for "Free Form Text Only". This group receives only about 61 approvals for every 100 White-reference approvals.

FLAGGED GROUPS

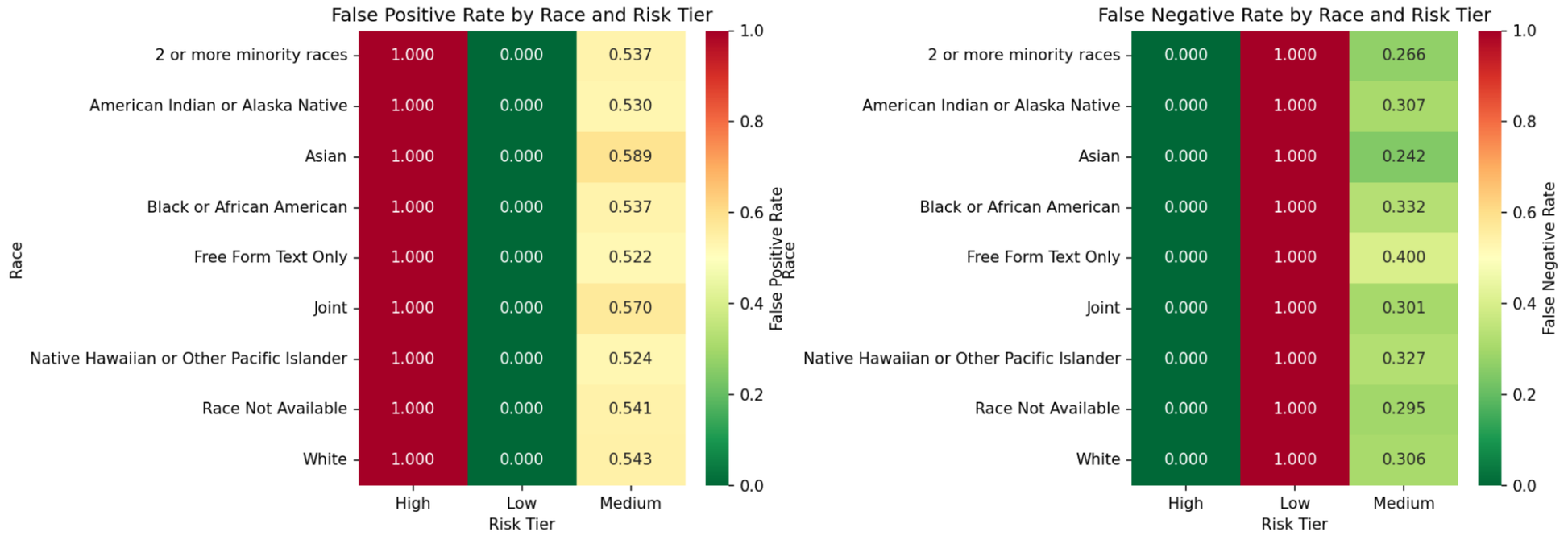
"Free Form Text Only"	AIR 0.606
Native Hawaiian / Pacific Islander	AIR 0.787
American Indian / Alaska Native	AIR 0.800



ERROR-RATE DISPARITIES BY RISK TIER

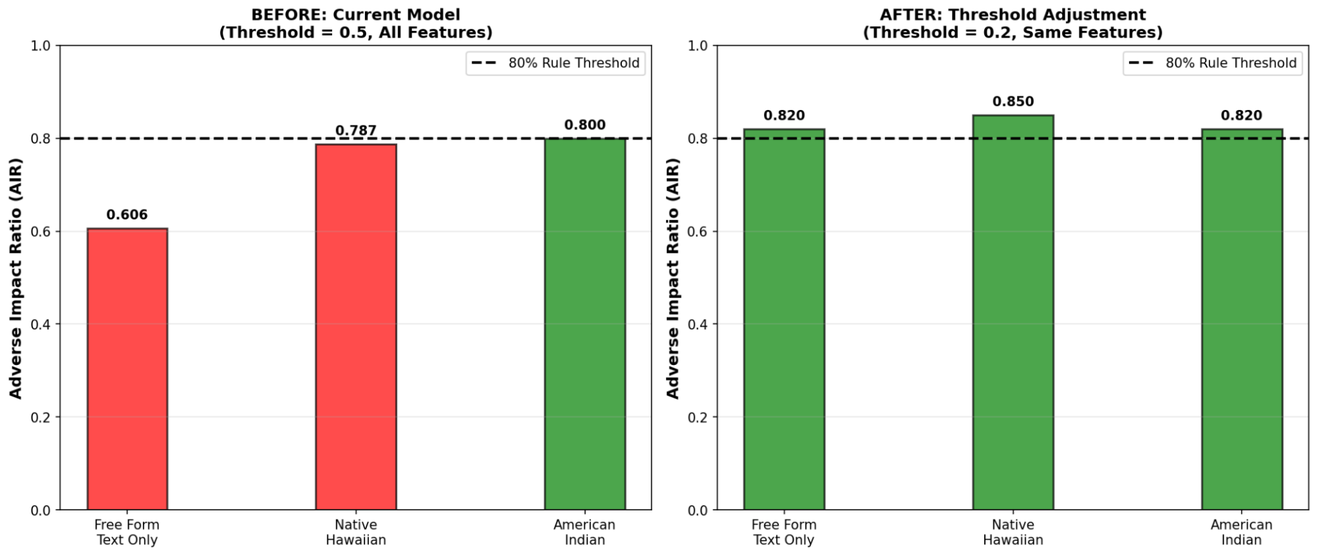
Average model quality hides subgroup-level behavior

Risk-tier and subgroup cuts show whether model errors are concentrated rather than evenly distributed.



BIAS MITIGATION EVIDENCE

Threshold adjustment improves AIR but increases lender risk



BEFORE → AFTER

0.606 → ≥0.820

Minimum AIR improves above 0.80

THRESHOLD

0.20

Fairness-pass threshold tested

COST

19.1% FPR

Lender risk increases

RESIDUAL RISKS AFTER MITIGATION

What remains even after the fairness fix?

An exemplary audit names residual risks and the conditions under which they are acceptable.

Reduced predictive power	Remaining correlated features	Data quality differences	Demographic drift	Intersectional harm
Threshold adjustment increases false approvals and lender credit risk.	Income, DTI, and geography may still encode historical inequality.	Applicant-supplied fields may vary by employment type or documentation access.	Applicant population may shift after deployment.	Race × sex combinations can remain disadvantaged even if aggregate AIR improves.
ACCEPT IF LOSSES STAY WITHIN RISK APPETITE.	NOT ACCEPTABLE LONG-TERM WITHOUT MONITORING.	ACCEPT ONLY WITH COMPLETENESS AUDITS.	RETRAIN IF DRIFT EXCEEDS TRIGGER.	QUARTERLY INTERSECTIONAL AUDIT REQUIRED.

DEPLOYMENT DEFENSIBILITY

Recommendation: deploy with conditions

VERDICT

DEPLOY WITH CONDITIONS

Not autonomous denial; use with governance.

BEFORE LAUNCH

Risk committee approval
Compliance/legal sign-off
Human review for borderline denials

MONITORING

Monthly AIR review
Quarterly demographic drift Annual retraining decision

AUDIT RECORD

GitHub repo as reproducible evidence
Versioned code, outputs, and model card

PULL / RE-EVALUATE TRIGGERS

AIR < 0.80 in monitoring
PSI > 0.25 or Wasserstein > 0.05
AUC drops below 0.75
Loan losses exceed risk appetite
Regulatory inquiry or civil-rights complaint

FINAL RECOMMENDATION

A justified model with documented risks

Responsible ML is not about claiming perfection; it is about accountability, transparency, and control.

APPROVE CONDITIONAL DEPLOYMENT ONLY IF:

01 Fairness-pass threshold is approved by risk/compliance

02 Monthly AIR monitoring and quarterly drift checks are active

03 Borderline denials receive human review

04 GitHub audit record is maintained and versioned

05 Shutdown/retraining triggers are enforced

VERDICT

Deploy with conditions

Pull if fairness, drift, performance, or regulatory triggers are breached.